

CALEB J. BALLARD

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EDUCATION

Master of Science, Software Engineering

University of Houston - Clear Lake, Engineering | GPA 3.9 | December 2026

Bachelor of Science, Computer Engineering

University of Houston, Cullen College of Engineering | GPA 3.5 | May 2022

SKILLS

Programming Languages: Python, C++, C

Tools & Platforms: GitHub, Visual Studio, Blueprint Scripting, DOORS, Linux, Agile, Unreal Engine 5, MATLAB

Certifications: ISS Command Certification - NASA

EXPERIENCE

Systems Test Engineer

KBR Inc, Houston, TX | September 2023 - Present

- Collaborating with NASA and international partners to capture system requirements and develop comprehensive test plans for Mission Control Center Systems (MCCS).
- Leading the creation and execution of detailed test & integration procedures to verify MCCS functionality for Gateway, ISS, Starliner, and Orion.
- Executing system risk evaluation and supportability assessments through comprehensive regression testing, ensuring high-quality system performance.
- Maintaining and updating test tools and scripts, leveraging DOORS for efficient project management and quality assurance.

Space Systems Operations Engineer

KBR Inc, Houston, TX | June 2022 - September 2023

- Gained deep technical expertise in 1553 architecture, audio/video, UHF, S-Band, and Ku-Band systems for the International Space Station (ISS).
- Executed real-time problem solving, responding to CDH and C&T system failures, and refining interdisciplinary coordination.
- Developed front-end portion of displays for Mission Control, enhancing operational efficiency and user experience.
- Collaborated on updating memory maps for Multiplexer Demultiplexers (MDM) on the ISS.

ACADEMIC PROJECTS

XR Solar System Observatory Experience – Unreal Engine 5

- Developed an interactive Solar System for Meta Quest 3 using OpenXR and Blueprint Scripting where planets are displayed on the ceiling and dynamically move to the user's view when selected.
- Integrated an AI-generated guide that provides detailed information about each planet, enhancing learning and engagement.
- Enabled users explore a close-up view of each planet's surface for an immersive visual experience.